

Daily Logic Drill #4 — Answers & Investigations

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Disproof by Counterexample · Boundary Case Discovery · Derivative & Function Counterexamples · Universal Refutation.

Section A

1. Provide the minimal positive integer counterexample n to: “For all positive integers n , if n is prime, then $n^2 + 2$ is not prime.” (after TMUA 2022 Paper 2 Q3)

Answer: 3

Method: Test small primes: For $n = 2$, $2^2 + 2 = 6$ (composite, not a counterexample). For $n = 3$, 3 is prime and $3^2 + 2 = 11$, which is prime! Thus $n = 3$ satisfies the premise but falsifies the conclusion, making it the minimal counterexample.

Investigate further: For any prime $p > 3$, $p \equiv \pm 1 \pmod{3}$, so $p^2 \equiv 1 \pmod{3}$, which means $p^2 + 2 \equiv 0 \pmod{3}$, so $p^2 + 2$ is always a multiple of 3 and thus composite. Hence $n = 3$ is the *only* prime counterexample!

2. Find the smallest positive integer a for which there exist positive integers b, c such that $a \mid bc$ but $a \nmid b$ and $a \nmid c$. (after TMUA 2021 Paper 2 Q4)

Answer: 4

Method: If a is prime, Euclid’s Lemma guarantees $a \mid bc \implies a \mid b \vee a \mid c$. Thus a must be composite. The smallest composite number is $a = 4$. Taking $b = 2$ and $c = 2$, we have $bc = 4$, so $4 \mid 4$, but $4 \nmid 2$. Thus $a = 4$ is the minimal counterexample.

Investigate further: If a must be squarefree, what is the smallest counterexample? $a = 6$, with $b = 2, c = 3$.

3. Provide a function $f(x)$ with $f'(x) > 0$ for all $x \in \mathbb{R}$ that serves as a counterexample to: “If $f'(x) > 0$ for all x , then $f(x) > 0$ for all x .” (after TMUA 2018 Paper 2 Q6)

Answer: “ $f(x) = x$ ”

Method: The linear function $f(x) = x$ has $f'(x) = 1 > 0$ for all $x \in \mathbb{R}$, but $f(x) \leq 0$ for all $x \leq 0$ (e.g. $f(-5) = -5 \not> 0$). Another valid counterexample is $f(x) = e^x - 2$.

Investigate further: A positive derivative guarantees that the function is strictly increasing, not that its values are positive.

4. Find a real number $x > 0$ that serves as a counterexample to: “ $x > \frac{1}{x}$ for all $x > 0$.”

Answer: $\frac{1}{2}$

Method: For $x = \frac{1}{2} > 0$, we have $\frac{1}{x} = 2$. Since $\frac{1}{2} < 2$, the inequality fails. Any $x \in (0, 1]$ disproves the claim.

Investigate further: The inequality $x > 1/x$ on positive reals holds if and only if $x > 1$.

5. Provide a pair of real numbers (x, y) that serves as a counterexample to: “For all $x, y \in \mathbb{R}$, if $x^2 > y^2$, then $x > y$.”

Answer: $(-3, 1)$

Method: Let $x = -3$ and $y = 1$. Then $x^2 = 9$ and $y^2 = 1$, so $x^2 > y^2$ holds. However, $-3 < 1$, so $x > y$ is false.

Investigate further: Squaring preserves inequality only when both numbers are non-negative.

6. Find the smallest positive integer n for which $2^n - 1$ is composite despite n being prime.

Answer: 11

Method: Primes n : $2^2 - 1 = 3$ (prime), $2^3 - 1 = 7$ (prime), $2^5 - 1 = 31$ (prime), $2^7 - 1 = 127$ (prime). For $n = 11$: $2^{11} - 1 = 2047 = 23 \times 89$, which is composite. Thus $n = 11$ is the smallest counterexample.

Investigate further: Mersenne numbers $2^p - 1$ need not be prime even when p is prime.

7. Provide a pair of non-negative real numbers (a, b) that serves as a counterexample to: “ $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$.”

Answer: $(1, 1)$

Method: For $a = 1, b = 1$, $\sqrt{a+b} = \sqrt{2}$, whereas $\sqrt{a} + \sqrt{b} = 1 + 1 = 2$. Since $\sqrt{2} \neq 2$, the identity fails.

Investigate further: The equality $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ holds if and only if $ab = 0$.

8. Find the smallest positive integer n that serves as a counterexample to: “ $n^2 + 1$ is always prime.”

Answer: 3

Method: For $n = 1$, $1^2 + 1 = 2$ (prime). For $n = 2$, $2^2 + 1 = 5$ (prime). For $n = 3$, $3^2 + 1 = 10 = 2 \times 5$, which is composite. Smallest n is 3.

Investigate further: For which other small integers is $n^2 + 1$ composite? $n = 5 \implies 26 = 2 \times 13$, $n = 7 \implies 50$.

9. Provide a differentiable function $f : [1, \infty) \rightarrow \mathbb{R}$ with $\lim_{x \rightarrow \infty} f'(x) = 0$ that disproves: “If $\lim_{x \rightarrow \infty} f'(x) = 0$, then $\lim_{x \rightarrow \infty} f(x)$ exists and is finite.”

Answer: “ $f(x) = \sqrt{x}$ ”

Method: For $f(x) = \sqrt{x}$, $f'(x) = \frac{1}{2\sqrt{x}} \rightarrow 0$ as $x \rightarrow \infty$, but $f(x) = \sqrt{x} \rightarrow \infty$. Another standard counterexample is $f(x) = \ln x$, where $f'(x) = 1/x \rightarrow 0$ while $\ln x \rightarrow \infty$.

Investigate further: A vanishing derivative means the growth rate slows to zero, but the total accumulated growth can still diverge to infinity.

10. Find the smallest odd composite integer n that is not a prime power.

Answer: 15

Method: Odd composites in order: $9 = 3^2$ (prime power), $15 = 3 \times 5$ (not a prime power). Thus 15 is the smallest.

Investigate further: Any non-prime-power odd composite has at least two distinct odd prime factors.

Section B

1. A student claims: “If a polynomial $f(x)$ satisfies $f(n) \in \mathbb{Z}$ for all $n \in \mathbb{Z}$, then $f'(n) \in \mathbb{Z}$ for all $n \in \mathbb{Z}$.” Provide a quadratic polynomial $f(x)$ that serves as a counterexample. (after TMUA 2017 Paper 2 Q16)

Answer: ” $f(x) = \frac{x(x-1)}{2}$ ”

Method: For $f(x) = \frac{x^2-x}{2} = \binom{x}{2}$, the product of two consecutive integers $n(n-1)$ is always even, so $f(n) \in \mathbb{Z}$ for all $n \in \mathbb{Z}$. However, $f'(x) = x - \frac{1}{2}$. For any integer n , $f'(n) = n - \frac{1}{2} \notin \mathbb{Z}$. This disproves the claim.

Investigate further: This counterexample relies on the binomial basis polynomials $\binom{x}{k}$, which are integer-valued on $\mathbb{Z}$ but do not have integer coefficients.

2. Give a pair of irrational numbers a, b whose sum is rational.

Answer: ” $a = \sqrt{2}, b = -\sqrt{2}$ ”

Method: $\sqrt{2}$ and $-\sqrt{2}$ are both irrational, but their sum is $\sqrt{2} + (-\sqrt{2}) = 0 \in \mathbb{Q}$.

Investigate further: Another pair: $a = 1 + \sqrt{3}, b = 1 - \sqrt{3}$, where $a + b = 2$.

3. Give a pair of irrational numbers a, b whose product is rational.

Answer: ” $a = \sqrt{2}, b = \sqrt{2}$ ”

Method: $\sqrt{2}$ is irrational, but $ab = \sqrt{2} \times \sqrt{2} = 2 \in \mathbb{Q}$.

Investigate further: A pair of distinct positive irrationals: $a = \sqrt{8}, b = \sqrt{2}$, where $ab = \sqrt{16} = 4$.

4. Provide a pair of functions $f, g : [0, \infty) \rightarrow \mathbb{R}$ such that $f(x) \geq g(x)$ for all $x \geq 0$, but $f'(0) < g'(0)$. (after TMUA 2021 Paper 2 Q12)

Answer: ” $f(x) = 10, g(x) = x$ ”

Method: For $x \in [0, 5]$, $f(x) = 10 \geq x = g(x)$. To make $f(x) \geq g(x)$ for all $x \geq 0$, let $f(x) = 10$ and $g(x) = 10 - e^{-x}$. Then $f(x) = 10 > g(x)$ for all $x \geq 0$, but $f'(0) = 0$ while $g'(0) = 1 > 0$, so $f'(0) < g'(0)$.

Investigate further: Inequalities between functions cannot be differentiated; a function can sit entirely above another while having a smaller derivative.

5. Find the smallest positive integer n that is a counterexample to: “ $n^2 + n + 41$ is prime for all positive integers n .”

Answer: 40

Method: For $n = 40$, $n^2 + n + 41 = 40^2 + 40 + 41 = 40(41) + 41 = 41(40 + 1) = 41^2 = 1681$, which is composite (41×41). For all $1 \leq n \leq 39$, $n^2 + n + 41$ is prime. Thus $n = 40$ is the minimal counterexample.

Investigate further: At $n = 41$, $41^2 + 41 + 41 = 41 \times 43$, which is also composite, but 40 is smaller.

6. Provide a counterexample of three finite sets A, B, C showing that $A \cup B = A \cup C$ does not imply $B = C$.

Answer: " $A = \{1\}, B = \{1\}, C = \emptyset$ "

Method: Here $A \cup B = \{1\} \cup \{1\} = \{1\}$, and $A \cup C = \{1\} \cup \emptyset = \{1\}$. The unions are equal, but $B = \{1\} \neq \emptyset = C$.

Investigate further: Union is not an invertible operation: elements absorbed by A lose their distinguishing presence in B or C .

7. Find the smallest positive integer n for which $3^n + 2$ is composite.

Answer: 5

Method: Test values of n : $n = 1 \implies 3^1 + 2 = 5$ (prime); $n = 2 \implies 3^2 + 2 = 11$ (prime); $n = 3 \implies 3^3 + 2 = 29$ (prime); $n = 4 \implies 3^4 + 2 = 83$ (prime); $n = 5 \implies 3^5 + 2 = 243 + 2 = 245 = 5 \times 49$ (composite). The smallest positive integer is $n = 5$.

Investigate further: Modulo 5: $3^1 \equiv 3, 3^2 \equiv 4, 3^3 \equiv 2, 3^4 \equiv 1$, so $3^5 \equiv 3 \implies 3^5 + 2 \equiv 5 \equiv 0 \pmod{5}$.

8. A student conjectures: "For every odd prime p , the integer $2^p - p$ is divisible by either 3 or 5." Find the smallest odd prime p that serves as a counterexample. (after TMUA 2021 Paper 2 Q10)

Answer: 7

Method: Test odd primes:

- $p = 3$: $2^3 - 3 = 8 - 3 = 5$ (divisible by 5).
- $p = 5$: $2^5 - 5 = 32 - 5 = 27$ (divisible by 3).
- $p = 7$: $2^7 - 7 = 128 - 7 = 121 = 11^2$. 121 is divisible by neither 3 nor 5.

Thus $p = 7$ is the smallest odd prime counterexample.

Investigate further: Small patterns in modular arithmetic often break precisely at $p = 7$, the first prime where neither $2^p \equiv p \pmod{3}$ nor $2^p \equiv p \pmod{5}$ holds.

9. Provide integers a, b and a modulus $m > 1$ showing that $a^2 \equiv b^2 \pmod{m}$ does not imply $a \equiv \pm b \pmod{m}$.

Answer: " $m = 8, a = 1, b = 3$ "

Method: For $m = 8, a = 1, b = 3$: $a^2 = 1 \equiv 1 \pmod{8}$ and $b^2 = 9 \equiv 1 \pmod{8}$, so $a^2 \equiv b^2 \pmod{8}$. However, $3 \not\equiv 1 \pmod{8}$ and $3 \not\equiv -1 \equiv 7 \pmod{8}$.

Investigate further: The failure occurs because 8 is composite; in $\mathbb{Z}/8\mathbb{Z}$, the polynomial $x^2 - 1$ has four roots $\{1, 3, 5, 7\}$.

10. Give a counterexample function $f : [0, 1] \rightarrow [0, 1]$ showing that a continuous function need not have a UNIQUE fixed point.

Answer: " $f(x) = x$ "

Method: The identity function $f(x) = x$ maps $[0, 1]$ continuously to $[0, 1]$. Every single $x \in [0, 1]$ is a fixed point, meaning there are infinitely many fixed points rather than a unique one.

Investigate further: Brouwer's Fixed Point Theorem guarantees the *existence* of at least one fixed point, but never uniqueness.

Section C

1. Consider the statement: " $(*)$ If a whole number n is 2 more or 4 more than a multiple of 5, then n is prime." How many counterexamples to $(*)$ are there in the range $1 \leq n \leq 50$? (after TMUA 2019 Paper 2 Q4)

Answer: 13

Method: Numbers of the form $5k + 2$ in $[1, 50]$ are: 2, 7, 12, 17, 22, 27, 32, 37, 42, 47. The composites among these are 12, 22, 27, 32, 42 (5 counterexamples).

Numbers of the form $5k + 4$ in $[1, 50]$ are: 4, 9, 14, 19, 24, 29, 34, 39, 44, 49. The composites among these are 4, 9, 14, 24, 34, 39, 44, 49 (8 counterexamples).

Total counterexamples = $5 + 8 = 13$.

Investigate further: Note that 2 and 7 satisfy the premise and are prime, so they are not counterexamples.

2. Which of the following values of n is a counterexample to the claim: "For all positive integers n , if n is prime, then $n^3 - n$ is NOT divisible by 24?" (after TMUA 2022 Paper 2 Q3)

Answer: E) Both $n = 5$ and $n = 7$.

Method: Factor $n^3 - n = (n - 1)n(n + 1)$. For any prime $p \geq 5$, p is odd, so $p - 1$ and $p + 1$ are consecutive even integers, one of which is divisible by 4, so their product is divisible by 8. Furthermore, among $p - 1, p, p + 1$, exactly one is divisible by 3; since $p \geq 5$ is prime, $3 \nmid p$, so $3 \mid (p - 1)(p + 1)$. Thus for any prime $p \geq 5$, $24 \mid (p^3 - p)$. Both $n = 5$ ($5^3 - 5 = 120 = 24 \times 5$) and $n = 7$ ($7^3 - 7 = 336 = 24 \times 14$) are divisible by 24, refuting the claim. Option E.

Investigate further: This means every single prime $p \geq 5$ is a counterexample to the claim!

3. A student claims: "If $f : \mathbb{R} \rightarrow \mathbb{R}$ is twice differentiable, $f(0) = 0$, and $f''(x) > 0$ for all $x \in \mathbb{R}$, then $f(x) \geq 0$ for all $x \geq 0$." Which of the following functions serves as a counterexample? (after TMUA 2018 Paper 2 Q6)

Answer: C) $f(x) = x^2 - 3x$.

Method: Check the conditions for $f(x) = x^2 - 3x$: $f(0) = 0$; $f'(x) = 2x - 3$; $f''(x) = 2 > 0$ for all x . All hypotheses hold. However, for $x \in (0, 3)$, $f(x) = x(x - 3) < 0$. For instance, $f(1) = -2 < 0$. This disproves the claim. Option C.

Investigate further: A positive second derivative means the graph is convex (bending upward), but if the initial slope $f'(0)$ is negative, the function dips below zero before rising.

4. Consider the statement: "For all positive integers a, b, c , if $a \mid bc$, then $a \mid b$ or $a \mid c$." Which of the following triples (a, b, c) is a valid counterexample? (after TMUA 2021 Paper 2 Q4)

Answer: B) (6, 4, 9).

Method: Check option B: $bc = 4 \times 9 = 36$. $a = 6$ divides 36 ($36/6 = 6$). However, $6 \nmid 4$ and $6 \nmid 9$. Both disjuncts fail while the premise holds. Thus (6, 4, 9) is a valid counterexample. In option A, $a = 2 \mid 4$; in C, $a = 5 \mid 10$; in D, $a = 7 \mid 14$.

Investigate further: A counterexample requires a to share distinct prime factors between b and c .

5. Which of the following polynomials satisfies $f(n) \in \mathbb{Z}$ for all $n \in \mathbb{Z}$, but $f'(0) \notin \mathbb{Z}$? (after TMUA 2017 Paper 2 Q16)

Answer: D) Both B and C.

Method: For B: $f(x) = \frac{x^2+x}{2} = \binom{x+1}{2}$, integer-valued on $\mathbb{Z}$. $f'(x) = x + \frac{1}{2} \implies f'(0) = \frac{1}{2} \notin \mathbb{Z}$.
For C: $f(x) = \frac{x^3-x}{6} = \frac{(x-1)x(x+1)}{6} = \binom{x+1}{3}$, integer-valued on $\mathbb{Z}$ since the product of three consecutive integers is always divisible by 6. $f'(x) = \frac{3x^2-1}{6} \implies f'(0) = -\frac{1}{6} \notin \mathbb{Z}$.
Both B and C satisfy the condition. Option D.

Investigate further: Every integer-valued polynomial is an integer linear combination of the binomial polynomials $\binom{x}{k}$.

6. A student conjectures: “If a continuous function $f : [-\pi, \pi] \rightarrow \mathbb{R}$ satisfies $\int_{-\pi}^{\pi} f(x)dx = 0$, then $f(x) = 0$ for all $x \in [-\pi, \pi]$.” Which of the following is a valid counterexample? (after TMUA 2019 Paper 2 Q14)

Answer: D) All of A, B, C.

Method: Evaluate the integral for each function on $[-\pi, \pi]$:

- A: $\int_{-\pi}^{\pi} \cos x \, dx = [\sin x]_{-\pi}^{\pi} = 0 - 0 = 0$, but $\cos 0 = 1 \neq 0$.
- B: $\int_{-\pi}^{\pi} \sin x \, dx = [-\cos x]_{-\pi}^{\pi} = 1 - 1 = 0$, but $\sin(\pi/2) = 1 \neq 0$.
- C: $\int_{-\pi}^{\pi} x \, dx = \left[\frac{x^2}{2}\right]_{-\pi}^{\pi} = 0$, but $f(1) = 1 \neq 0$.

All three functions have zero integral yet are not identically zero. Option D.

Investigate further: Any non-trivial odd function on a symmetric interval $[-a, a]$ integrates to zero.

7. Consider two statements about functions $f, g : [0, \infty) \rightarrow \mathbb{R}$ with $f(0) = g(0) = 0$:

- $f'(x) \geq g'(x)$ for all $x \geq 0 \implies f(x) \geq g(x)$ for all $x \geq 0$.
- $f(x) \geq g(x)$ for all $x \geq 0 \implies f'(x) \geq g'(x)$ for all $x \geq 0$.

Which of I and II are true? (after TMUA 2021 Paper 2 Q12)

Answer: B) I only.

Method: By the Fundamental Theorem of Calculus, $f(x) - g(x) = \int_0^x (f'(t) - g'(t)) \, dt$. If $f'(t) \geq g'(t)$, the integrand is non-negative, so the integral is ≥ 0 , proving $f(x) \geq g(x)$. Thus statement I is TRUE. For statement II: consider $f(x) = 2x$ and $g(x) = x + \sin x$. For $x \geq 0$, $f(x) - g(x) = x - \sin x \geq 0$, so $f(x) \geq g(x)$. However, $f'(x) = 2$ and $g'(x) = 1 + \cos x$. At $x = 0$, $f'(0) = 2 \geq 2$, but consider $f(x) = 10x$ and $g(x) = 10x - 5 \cos(10x) + 5$: oscillating derivatives frequently exceed $f'(x)$. Statement II is FALSE. Option B.

Investigate further: Integration preserves inequalities; differentiation does not.

8. Consider the theorem: “An odd prime p can be written as the sum of two integer squares if and only if $p \equiv 1 \pmod{4}$.” Which of the following would constitute a counterexample to the **converse** of this theorem? (after *TMUA 2018 Paper 2 Q17*)

Answer: A) A prime $p \equiv 1 \pmod{4}$ that cannot be written as the sum of two squares.

Method: The theorem states $P \iff Q$, where P : “ p is a sum of two squares” and Q : “ $p \equiv 1 \pmod{4}$ ” (for odd primes). The converse is $Q \Rightarrow P$: “If an odd prime $p \equiv 1 \pmod{4}$, then p can be written as the sum of two squares.” A counterexample to $Q \Rightarrow P$ must satisfy the hypothesis Q ($p \equiv 1 \pmod{4}$) but fail the conclusion P (cannot be written as the sum of two squares). Option A.

Investigate further: Refuting a conditional statement requires satisfying the antecedent while falsifying the consequent.

Section D

1. Consider: I. “ $n = 41$ is a counterexample to ‘ $n^2 + n + 41$ is always prime’.” II. “ $n = 40$ is the smallest counterexample to ‘ $n^2 + n + 41$ is always prime’.” III. “ $n = 4$ is a counterexample to ‘ $n^2 + n + 41$ is always prime’.” Which of I, II, III are true?

Answer: E) I and II only

Method: I: $41^2 + 41 + 41 = 41(41 + 1 + 1) = 41 \times 43$, composite. So I is true.

II: For $1 \leq n \leq 39$, $n^2 + n + 41$ is prime. At $n = 40$, $40^2 + 40 + 41 = 41^2$, composite. Thus $n = 40$ is indeed the smallest positive integer counterexample. II is true.

III: For $n = 4$, $4^2 + 4 + 41 = 61$, which is prime. So $n = 4$ is not a counterexample. III is false.

Hence, I and II only are true.

Investigate further: Euler’s polynomial $n^2 + n + 41$ is the classical demonstration that finite verification cannot prove universal claims.

2. Prove by contradiction: “There do not exist positive integers a, b with $a^2 - b^2 = 1$.”

Answer: Proof by contradiction

Method: Assume toward a contradiction that there exist positive integers a, b such that $a^2 - b^2 = 1$. Factorising as a difference of two squares gives $(a - b)(a + b) = 1$. Since $a, b \in \mathbb{Z}^+$, we have $a \geq 1$ and $b \geq 1$, so $a + b \geq 2$. However, the only factorisation of 1 into positive integers is 1×1 , requiring $a + b = 1$. This contradicts $a + b \geq 2$. Thus, no such positive integers exist.

Investigate further: Do non-zero integer solutions exist if a, b are not restricted to positive integers?

3. A student claims: “For every positive integer n , at least one of $n, n + 2, n + 4$ is divisible by 3.” Determine, with proof, whether this claim is true.

Answer: True

Method: Proof: Any positive integer n can be written as $3k, 3k + 1$, or $3k + 2$ for some integer $k \geq 0$. Case 1: If $n = 3k$, then n is divisible by 3.

Case 2: If $n = 3k + 1$, then $n + 2 = (3k + 1) + 2 = 3k + 3 = 3(k + 1)$, which is divisible by 3.

Case 3: If $n = 3k + 2$, then $n + 4 = (3k + 2) + 4 = 3k + 6 = 3(k + 2)$, which is divisible by 3.

In all cases, at least one of $n, n + 2, n + 4$ is divisible by 3. The claim is true.

Investigate further: Notice that $n, n+2, n+4 \equiv n, n-1, n+1 \pmod{3}$, so they cover all three residue classes modulo 3.

4. Give a counterexample to: “If p is prime, then $2^p + 1$ is prime.”

Answer: 3

Method: For $p = 3$ (which is prime), $2^p + 1 = 2^3 + 1 = 8 + 1 = 9 = 3^2$, which is composite. (Also for $p = 5$, $2^5 + 1 = 33 = 3 \times 11$).

Investigate further: For any odd prime p , $2^p + 1 = (2 + 1)(2^{p-1} - 2^{p-2} + \dots + 1) = 3 \times M$, so $2^p + 1$ is always a multiple of 3, and hence composite for all $p > 1$ with $2^p + 1 > 3$!

5. Statement (*): “For every positive integer $n \geq 2$, there is a prime strictly between n and $2n$.”
(a) Verify (*) for $n = 2, 3, 4, 5$ by direct computation. (b) Does this verification constitute a proof of (*)? Justify your answer.

Answer: (a) $n = 2 : 3$; $n = 3 : 5$; $n = 4 : 5$ or 7 ; $n = 5 : 7$. (b) No, finite checking does not prove a universal statement.

Method: (a) For $n = 2$, prime $3 \in (2, 4)$. For $n = 3$, prime $5 \in (3, 6)$. For $n = 4$, prime $5 \in (4, 8)$. For $n = 5$, prime $7 \in (5, 10)$.

(b) No. Verification for finitely many cases ($n = 2, 3, 4, 5$) does not constitute a proof for infinitely many positive integers $n \geq 2$. A single unchecked value could be a counterexample. (The statement is true for all $n \geq 2$, known as Bertrand’s Postulate, but proving it requires a general argument).

Investigate further: Bertrand’s Postulate was first proved by Chebyshev in 1852 using properties of the gamma function and prime factorisations of central binomial coefficients.

Top 5 Patterns Today

- Refuting Implications by Finding Witness Instances.** To disprove $\forall x, (P(x) \Rightarrow Q(x))$, find a single counterexample where $P(x)$ holds while $Q(x)$ fails. (see A1, A2, A3, A4, B8)
- Derivative and Function Inversion Traps.** A function satisfying $f'(x) > 0$ everywhere does not need to be positive ($f(x) = x$); similarly, $f(x) \geq g(x)$ does not force $f'(x) \geq g'(x)$. (see A3, A9, B4, C3, C7)
- Integer-Valued Polynomial Derivatives.** A polynomial integer-valued on $\mathbb{Z}$ (such as binomial coefficients $\binom{x}{2} = \frac{x^2-x}{2}$) need not have integer coefficients or integer derivatives. (see B1, C5)
- Prime Generator Boundaries and Divisibility.** Conjectures asserting primality (like $n^2 + n + 41$ or $2^p - p$) fail at structural boundaries ($n = 40, p = 7$) where algebraic factoring or non-residues intervene. (see A6, A8, B5, B7, D1)
- Non-Trivial Modular Residue Square Roots.** In non-prime moduli (like $m = 8$), the congruence $x^2 \equiv 1 \pmod{8}$ has four distinct roots ($\pm 1, \pm 3$), breaking the field property $a^2 \equiv b^2 \implies a \equiv \pm b$. (see A10, B9, C4)

Common Traps to Avoid

- Testing Only Positive or Symmetrical Values.** Order relations like $x^2 > y^2 \implies x > y$ fail only when considering negative values ($x = -3, y = 1$); always check signed edge cases. (see

A4, A5, A7, B2)

- 2. Confusing Implication Refutation with Converse Refutation.** Refuting the converse of $P \iff Q$ requires finding a witness satisfying Q but not P , rather than attempting to falsify the forward direction. *(see C8, A1, B8)*
- 3. Assuming Derivatives Inherit Function Inequalities.** The height of a curve $f(x) \geq g(x)$ provides zero information about its slope $f'(x) \geq g'(x)$; slopes can cross while functions remain separated. *(see B4, C3, C7)*
- 4. Assuming Finite Verification Equals Proof.** Checking that a statement holds for hundreds of small cases (Bertrand's Postulate, Euler's polynomial) demonstrates only the absence of small counterexamples, never a proof. *(see D5, B5, D1)*
- 5. Overlooking Composite Prime Factors $p \leq \sqrt{n}$.** Refuting that $a \mid bc \implies a \mid b \vee a \mid c$ requires a composite modulus a that splits its prime factors across b and c . *(see A2, C4, B9)*

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

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